
Web Application Testing

Learning Outcomes

After Completing this session, you should be able to explain about

- Common Testing Terminologies

- Performing, Unit, Integration, End-To-End Testing

- Functional and Non Functional Testing

- Regression and Performance Testing

- UI and REST API Testing

Question

What could go wrong if you do not test your application?

<https://www.information-age.com/the-10-worst-web-application-failures-294446/>

Why is Testing Necessary?

To evaluate work product such as requirement, design, and code

To verify whether all the requirements specified have been fulfilled

To validate whether the test object is complete and works as users and other stakeholders expect

Why is Testing Necessary?

To build confidence in the level of quality of the test object

To prevent defects

To find failures and defects

To reduce the level of risk of inadequate software quality

Attributes of Testing

There are three attributes of automated testing:

Scope - How much of the code does the test touch (Test Coverage)?

Tests can run on a single method, across the entire application, or somewhere in between.

Speed - How fast does the test run?

Test speeds can vary from millisecond to several minutes.

Attributes of Testing

There are three attributes of automated testing:

Fidelity - How "real-world" is the test?

For example, if part of the code you're testing needs to make a network request, does the test code actually make this network request or does it fake the result?

If the test actually talks with the network, this means it has higher fidelity, but also takes longer and could give flaky results if the network is occasionally down.

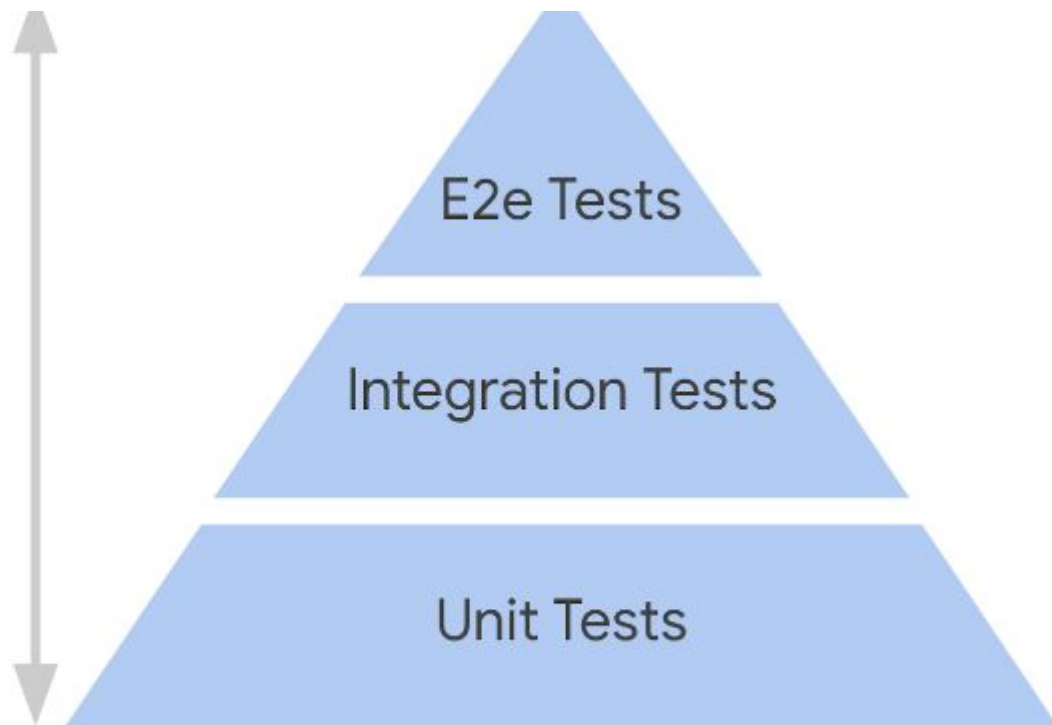
Attributes of Testing

Speed and **fidelity** are a trade off - the faster the test, generally the less fidelity and vice versa

The Testing Pyramid

High fidelity, large
scope, long running

Low fidelity, small
scope, very fast



of tests

The Testing Pyramid

Based on **scope**, **speed** and **fidelity**, we can have three levels of test

Unit Tests/Small Tests

These are highly focused tests that **run on a single class**, usually **a single method** in that class.

They **have low fidelity** since in the real world, your app involves much more than the execution of one method or class.

They **should be fast enough** to run every time you change your code.

The Testing Pyramid

Integration Tests/Medium Tests

These **test the interaction of several classes** to make sure they behave as expected when used together

One way to structure integration tests is to have them **test a single feature**

They **test a larger scope of code than unit tests** but are still optimized to run fast versus having full fidelity

The Testing Pyramid

End-to-End Tests/Large Tests

Test a **combination of features working together**

They **test large portions of the app**, simulate real usage closely and therefore are slow.

They **have the highest fidelity** and tell you that your application actually works as a whole.

Testing Terminology

Test Coverage

The **percentage of your code that is executed by your tests**.

If you have 100 lines of code, and your tests run through 80 of them, then you have 80% coverage.

Testing Terminology

Test Driven Development (TDD)

A school of programming thought that says instead of writing your feature code first, **you write your tests first**.

Then you write your feature code with the goal of passing your tests

Testing Terminology

Test Doubles

Test doubles are **objects that stand in for a real object**, such as a networking class and database class

when testing, you can **swap in a fake networking or database class to provide speed and determinism at the expense of fidelity**

Categories of **test doubles** include **Fakes**, **Dummies**, **Mocks**, and **Spies**

Testing Terminology

Given, When, Then

Also known as "**Arrange, Act, Assert.**"

They specify how test codes are grouped into sections

Testing Terminology

Given

The first section **sets up the desired state of the app**

When

This section of the test **specifies the functionality or actions that take place.**

Then

This is where you put all your statements that **verify what you expected to happen came to pass.**

Testing Terminology

White Box Testing

White box testing refers to testing where **we do have access to the source code** and **are able to inspect it**

Testing Terminology

Black Box Testing

In case of black box testing we **only have access to the tested artifact's external interface**

Question

What Kind of Test should you perform

Before, during, and after development?

Types of Test

Functional and **nonfunctional** testing

Functional testing answers the questions

Does the software do what it was intended to do?

Does it not do what it was not intended to do?

Types of Test

Functional and **nonfunctional** testing

Nonfunctional testing

targets a solution's quality attributes such as **usability, reliability, performance, maintainability, portability** and **security**.

Types of Test

Functional and **nonfunctional** testing

Functional tests target the **what**, whereas **non-functional** tests target the **how**

Performance Testing

Performance testing focuses on a system's **responsiveness**, **throughput**, and **reliability** given different loads

How fast does a web page load?

If a user clicks a button on the screen, are the contents immediately updated?

How long does it take to process 10,000 operations

Performance Testing

Load testing

The purpose of load testing is to determine the system's behavior in response to increased load

When the load is increased **beyond** the **maximum "normal load,"** load testing turns into **stress testing**

A special type of stress testing is **spike testing**, where the maximum normal load is exceeded very rapidly, as if there were a spike in the load

Security Testing

Testing Confidentiality

Data confidentiality and Privacy

Testing Integrity

Data and System Integrity

Testing Availability

Resources are available to authorized users and denied to others

Regression Testing

The purpose of regression testing is to **establish whether changes to the system have broken existing functionality** or caused old defects to resurface

Smoke Testing

The term smoke testing originated from engineers testing pipes by blowing smoke into them

If there was a crack, the smoke would seep out through it.

Smoke Testing

In software development, smoke testing refers to one or a few simple tests executed immediately after the system has been deployed.

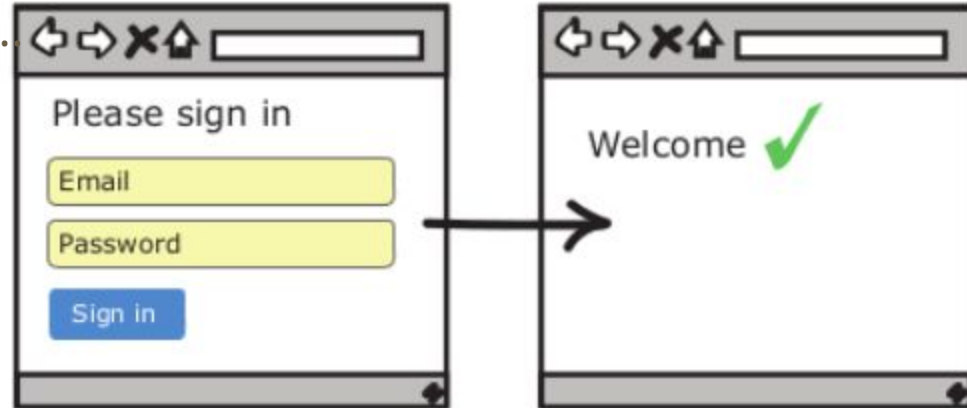
Smoke Testing

The “Hello World” of smoke testing is **logging into the application**, such a test provides a great deal of information

The application has been deployed successfully

The network connection works

The database could be reached, ...



Testing REST API: What to Test?

Security

testing checks how well the API is protected from malicious actors

it ensures that resources (data) are protected and only provided to authenticated or authorized clients

Testing REST API: What to Test?

Performance

testing makes sure that an API can respond to requests of specific sizes, or can respond to clients quickly enough to satisfy system requirements

Testing REST API: What to Test?

Other

Http Status Code

Response Header

Response Body

Testing UI

Write tests that work with browsers, and verify that the UI code is functioning properly

You can use tools such as

Playwright

Cypress

Selenium



Testing in NestJS

<https://docs.nestjs.com/fundamentals/testing>

Reference

Developer Testing: Building Quality into Software, By Alexander Tarlinder, 2017

Foundation of Software Testing, ISTQB Certification, 4th Edition

The Art of Software Testing, 3rd Edition